

SE LAB-3

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Architecture Selection:

We chose Layered Architecture for the Podcast Guest Scheduling and Outline Builder system. The system is organized into four layers: a Presentation Layer (booking portal used by guests and hosts), a Business Layer (scheduling rules, outline compilation), a Data Layer (encrypted storage of bookings, outlines and guest information), and an Integration Layer that isolates calls to the external Calendar Service, Notification/Email Service and PDF Production Sheet Engine identified in the Lab 1 use-case model.

Reason 1 — Matches the system's natural separation of concerns

Lab 1 already separates the domain into distinct concerns: guest-facing booking (FR-001, FR-004), host-side calendar control (FR-002), and production-sheet generation (FR-005), each depending on a different external system (Calendar Service, Notification/Email Service, PDF Production Sheet Engine). Layered architecture maps directly onto this: the Integration Layer absorbs all third-party API variability so the Business Layer's scheduling and outline logic never has to change when an external provider changes.

Reason 2 — Right-sized for a single-team, moderate-scale booking tool

The system has one primary workflow (book → notify → outline → compile PDF) and modest scale (a single host's episodes and guest list), so the independent-scaling benefits of microservices are not needed and would add operational overhead the team cannot justify. A single client-server split, on the other hand, would force calendar sync, email, encryption and PDF logic into one undifferentiated server layer, making NFR-001 and NFR-002 harder to enforce consistently. Layering keeps the codebase simple while still giving each concern its own boundary.

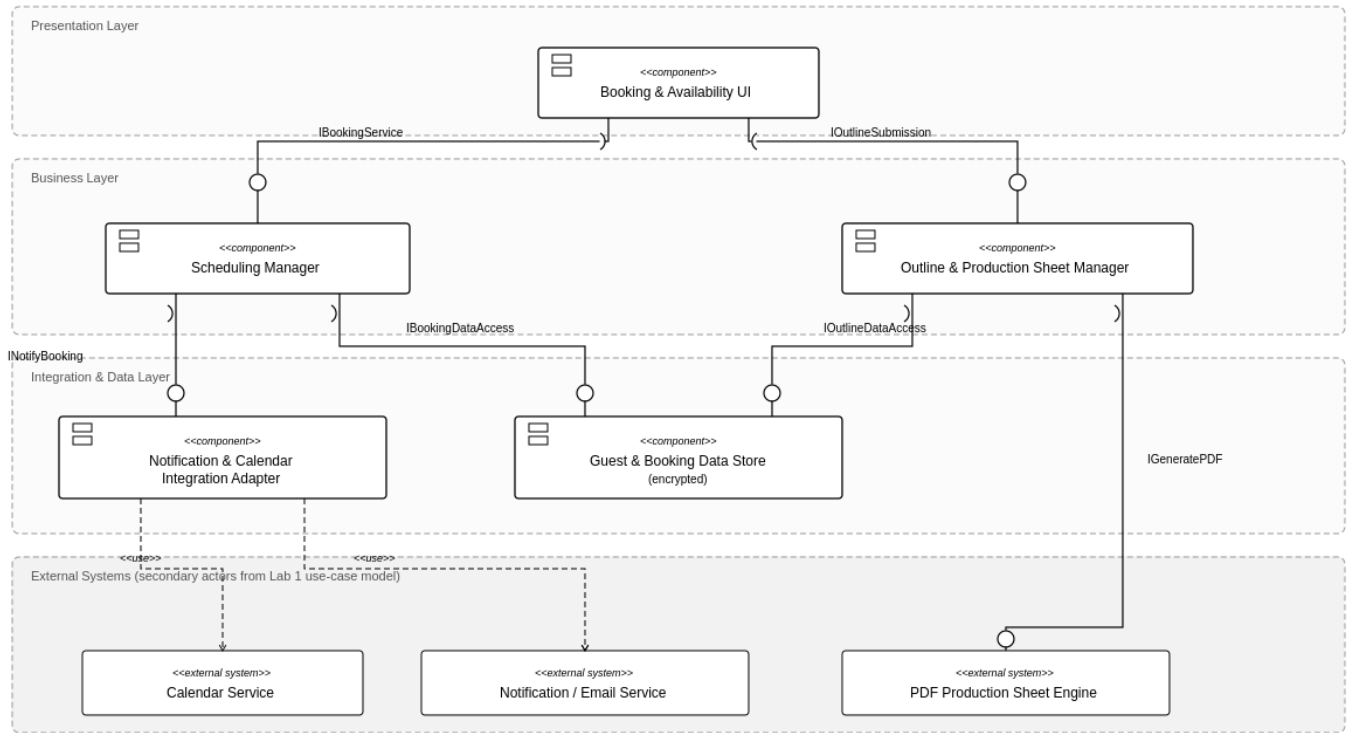
Security Advantage

All guest personal data (bio link, email — NFR-001) is encrypted and decrypted exclusively inside the Data Layer. The Presentation Layer and Business Layer only ever handle data through the Data Layer's interface, so raw guest information is never exposed to the UI or to the external Integration Layer, which reduces the system's attack surface and gives the team one place to audit and upgrade encryption.

Performance Benefit

Keeping the Outline & Production Sheet Manager and the PDF Production Sheet Engine in their own layer, separate from the request-handling Presentation Layer, lets outline data be pre-compiled in the Business Layer before the PDF call is made. This satisfies the sub-1-second PDF generation target in NFR-001 while keeping the Presentation Layer thin enough to meet the 2-second mobile page-load target in NFR-002.

Component Diagram - Podcast Guest Scheduling and Outline Builder



Legend

- Provided interface (ball) - placed on the component that offers the service
-) Required interface (socket) - placed on the component that needs the service
- Solid connector - internal service call, labelled with the interface name
- - -> Dashed <<use>> arrow - dependency on an external system
- Primary actors: Podcast Guest, Show Host - both interact only through the Booking & Availability UI.